LESSON PLAN

AIR NAVIGATION (21)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	17
Total No. of Lesson Plans	18	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Practical on different wire locking techniques used during maintenance.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Practical on different wire locking techniques used during maintenance.
5. Specific Objectives	21.1.10.1 Trainee will be able to explain how to carry out DI / DR on aircraft system.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration and practice of DI/DR wire locking techniques using dummy harnesses; Safety: Ensure power off, wear PPE; Supervision: Instructor monitors knot tightness and tool usage.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary purpose of wire locking in aircraft maintenance?
(b) Name two common types of wire locking techniques used in aviation.
(c) Why is it critical to inspect wire locks before flight?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Introduction to DI and DR wire locking • Tools required for wire locking • Step-by-step DI wire locking procedure • Step-by-step DR wire locking procedure • Common errors in wire locking	L & Ds	PPT, IB	15 min
(b)	• Inspection criteria for wire locks • Safety precautions during wire locking • Troubleshooting loose or damaged locks • Documentation of wire locking work • Quality assurance checks	L & Ds	PPT, PDF	20 min
(c)	• Final verification of wire locking • Cleaning and storage of tools • Review of DI/DR procedures • Addressing trainee queries • Preparation for next lesson	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between DI and DR wire locking techniques.
(b) Describe the steps to verify a correctly installed wire lock.

12. Home Assignment and Follow Up	Prepare a brief report on the use of advanced test equipment (Digital multimeter, Megger, Bonding tester) as covered in LP 18.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Training on use of advanced test equipment: - Digital & analog multimeter & Megger, Digital & analog Insulation tester, Bonding tester and No volt tester.	Refer to the master syllabus reference list	LP 18

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min